The logistic equation

Now that we understand how to find and classify solutions for an autonomous differential equation, let's look at a specific application of autonomous equations.

Autonomous equations are often used to model population growth over time, so let's start with a short review of simple exponential population growth.

Exponential population growth

A population that grows exponentially over time can be modeled by

$$P(t) = P_0 e^{kt}$$

where $P(t)$ is the population after time t, P_0 is the original population when time $t = 0$, and k is the growth constant.

Exponential growth is usually observed in smaller populations that aren't yet limited by their environment or the resources around them. In other words, this exponential equation is most useful for modeling a small population where growth is unconstrained.

There are four unknowns in this exponential equation, P, P_0, k, and t, so we'll need to know the values of three of these in order to find the value of the fourth. Let's try an example where we solve for a specific time t.

Example

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A bacteria population increases tenfold in 8 hours. Assuming exponential growth, how long did it take for the population to double?

We've been told that the population grows to 10 times its original size. If we call its original size P_0, then we can say that it grows to $10P_0$ after 8 hours. Substituting these three values into the exponential equation gives

$$10P_0 = P_0 e^{k(8)}$$

$$\frac{10P_0}{P_0} = e^{k(8)}$$

$$10 = e^{8k}$$

To solve for k, we'll apply the natural log to both sides of the equation, cancelling the $\ln e$, and then rearrange.

$$\ln 10 = \ln e^{8k}$$

$$\ln 10 = 8k$$

$$k = \frac{\ln 10}{8}$$

Now that we have a value for the growth constant k, we can can figure out how long it took for the population to double. If we say that P_0 is the original population, and $2P_0$ is double the original population, then the exponential equation gives

$$2P_0 = P_0 e^{\frac{\ln 10}{8}t}$$

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$$\frac{2P_0}{P_0} = e^{\frac{\ln 10}{8}t}$$

$$2 = e^{\frac{\ln 10}{8}t}$$

Apply the natural log to both sides of the equation, cancelling the $\ln e$, and then rearrange.

$$\ln 2 = \ln e^{\frac{\ln 10}{8}t}$$

$$\ln 2 = \frac{\ln 10}{8}t$$

$$t = \frac{8 \ln 2}{\ln 10}$$

$$t \approx 2.41$$

This result tells us that the bacteria doubled the size of its original population in about 2.41 hours, or about 2 hours and 25 minutes.

The logistic growth equation

The exponential equation $P(t) = P_0 e^{kt}$ is somewhat useful for modeling population, but its usefulness is limited. That's because populations don't tend to grow exponentially forever without bound.

Instead, populations are more likely to grow slowly at first when the population is small enough that replication is slow. As the population grows, the replication rate increases, and it seems like growth is increasing

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exponentially. But eventually, the population becomes large enough that the environment starts to constrain its growth, as it runs out of resources like space, energy, food, water, etc.

This slow-fast-slow growth pattern is modeled more effectively by the **logistic growth equation**

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{M}\right)$$

than it is by the exponential growth equation. In this logistic equation, dP/dt is the rate of growth of the population P over time t, k is the same growth constant that we had in the exponential growth equation, and M is the **carrying capacity** or saturation level, which is the largest population that can be sustained by the surrounding environment.

What we should notice is that this logistic growth equation is an autonomous first order differential equation, because we can see that there's no independent variable t on the right side of the logistic equation. The independent variable t only appears as part of the derivative dP/dt. So the right side of the logistic equation is a function in P only.

Which means we could name $f(P) = kP(1 - P/M)$, and then rewrite this logistic equation as

$$\frac{dP}{dt} = f(P)$$

As we saw before, we can find zeros by setting $f(P) = 0$.

$$kP\left(1 - \frac{P}{M}\right) = 0$$

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This equation will be true when

$$kP = 0$$

$$P = 0$$

or

$$1 - \frac{P}{M} = 0$$

$$\frac{P}{M} = 1$$

$$P = M$$

So the autonomous equation $dP/dt = f(P)$ will have equilibrium solutions when the population is 0, and when the population is at its carrying capacity. And this will always be the case for the logistic growth equation: the population will be stable when $P = 0$ and when $P = M$.

This should make sense. After all, the population won't increase or decrease if $P = 0$, because there are no members of the population to reproduce or die off. So a population at $P = 0$ will remain at $P = 0$ forever.

Similarly, the population won't increase or decrease at $P = M$, because this is the point at which the population is perfectly in balance with its environment. There aren't enough resources to support a larger population, so the population will not increase, but there are enough resources to support this exact population, so it also won't decrease either.

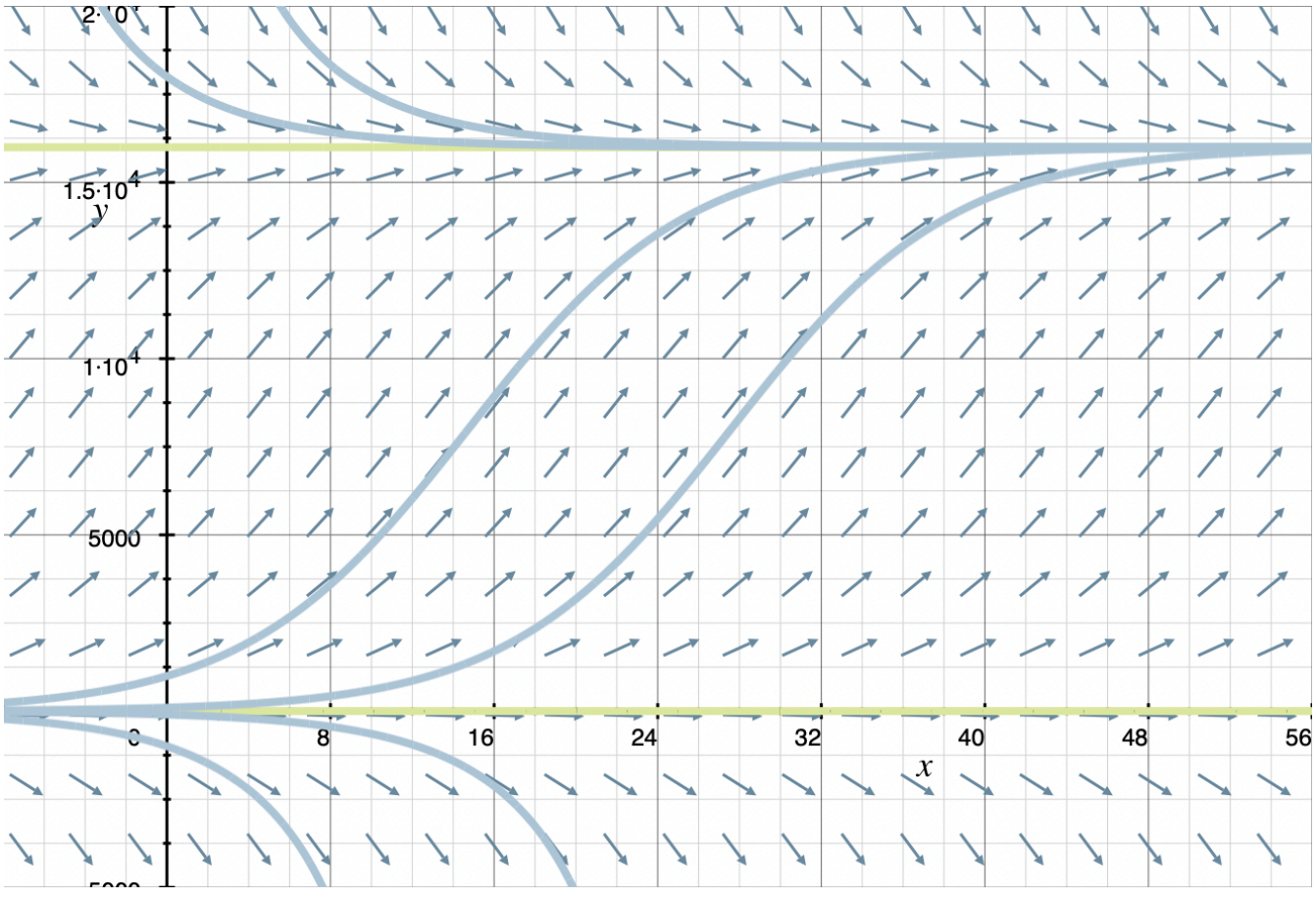
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Between $P = 0$ and $P = M$, the population is always increasing. As we mentioned earlier, the growth between $P = 0$ and $P = M$ follows a slow-fast-slow pattern (slow near $P = 0$, then faster, then slow again as we approach $P = M$).

If a population is ever greater than its carrying capacity, then growth will be negative until the population shrinks back to the level of carrying capacity. And while this doesn't apply in a real-world population model, the direction field of a logistic equation also shows that solution curves below $P = 0$ move away from the equilibrium solution.

Below is an example of the direction field for a logistic equation in which the carrying capacity is $M = 16{,}000$. It shows the equilibrium solutions at $P = 0$ and $P = M = 16{,}000$, as well as a few solution curves in each of the three intervals ($P < 0$, $0 < P < M$, and $M < P$).



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From the direction field, we can clearly see that $P = 0$ will always be an unstable equilibrium solution (because solution curves move away from it), while $P = M$ will always be a stable solution of the logistic equation (because solution curves move toward it).

Let's do an example with a population whose growth we'll model with the logistic equation.

Example

Describe the growth pattern of a population of ducks, assuming their population is modeled by the logistic growth equation, if the carrying capacity of their habitat is $2{,}500$ ducks.

Between $P = 0$ and $P = M = 2{,}500$, the duck population always increasing. In other words, in the interval $0 < P < 2{,}500$, the duck population will grow slowly when P is close to 0, then growth will speed up, and then the growth rate will level off again as P gets close to $2{,}500$.

If the population starts at a size of exactly 0, of course it will neither increase nor decrease, as indicated by the equilibrium solution at $P = 0$.

If the population starts at a size of exactly $2{,}500$, then it will neither increase nor decrease, as indicated by the equilibrium solution at $P = 2{,}500$. And if the duck population starts at a size greater than $2{,}500$, it will die off until it reaches stability at a size of $2{,}500$, as indicated by the stable equilibrium solution at $P = 2{,}500$.

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